BOARDING PASS | FOAI PROJECT

AI-Powered Personalized Travel Planning System

An explainable itinerary generator using rule-based AI

PASSENGERS

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ROUTE

Preferences to plan

101 destinations
8 scoring rules

Our journey today

STOP 01



Departure

The problem, our goal and objectives

STOP 02



Design

Why rules, architecture, inputs and knowledge base

STOP 03



Engine

Scoring rules, weights and budget logic

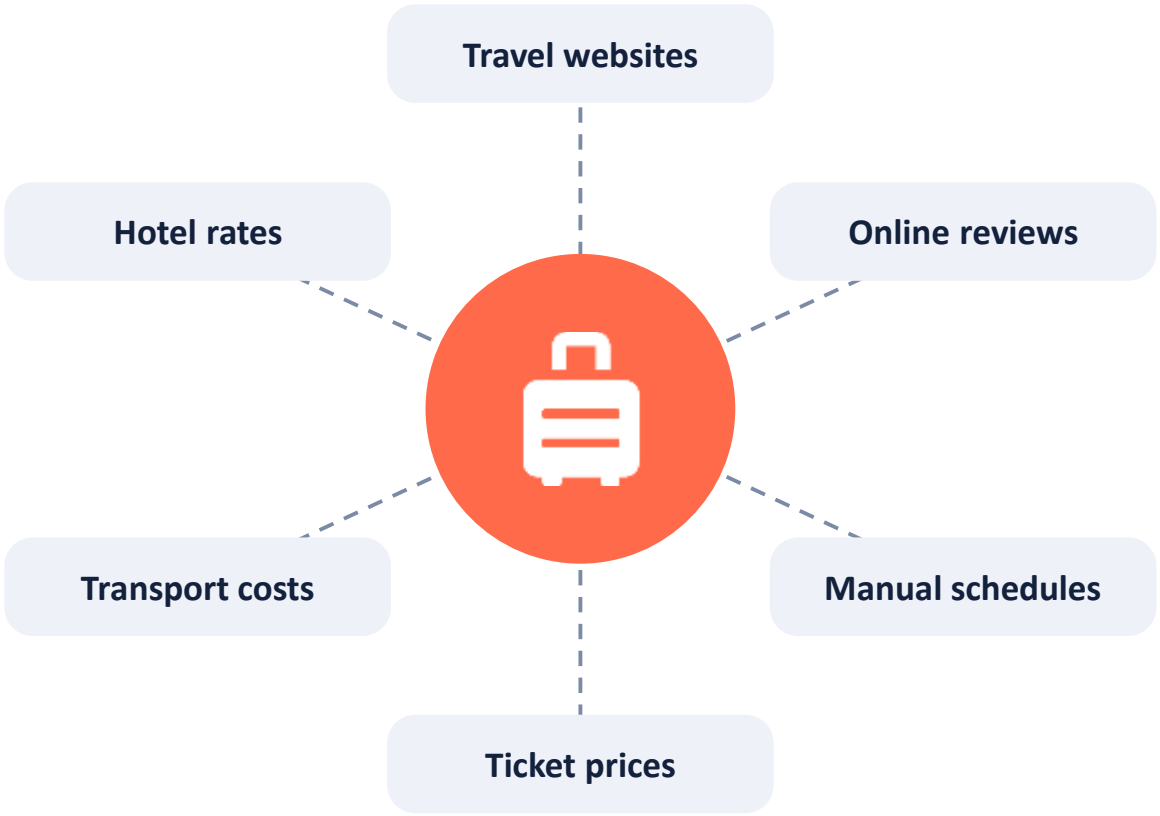
STOP 04



Arrival

Itinerary, explainability, stack and evaluation

Planning a trip means juggling too many sources



THE PAIN

- Time-consuming comparison
- Itineraries that are costly or unbalanced
- Generic lists ignore interests, time and budget

Our answer: one rule-based workflow from preferences to plan

Seven objectives, one goal

1. Collect destination, duration, budget, travellers and interests
2. Maintain a structured knowledge base
3. Recommend using predefined rules
4. Arrange a day-wise itinerary
5. Estimate transport, stay, food and entry fees
6. Flag budget overruns and suggest cheaper options
7. Explain results and test with defined cases

Plan smarter, and be able to say why

Why rules instead of machine learning?

ONE RULE, ONE REASON

IF interest is history

THEN prioritise museums, monuments and heritage sites

No training. No dataset.

	Rule-based (ours)	Typical ML
Training data	None needed	Large historical dataset
Explainability	Each pick traces to a rule	Often opaque
Changing behaviour	Edit a rule	Usually retrain
Best suited to	Small academic apps	Large-scale services

Seven stages, one decision-maker



Only stage 04 makes decisions. Every other stage prepares data or presents results.

What the traveller tells us

Trip basics

- Origin city
- Start date
- Trip length in days
- Number of travellers
- Solo, couple, family or group

Money and comfort

- Total budget in rupees
- Stay: hostel, 3-star, homestay, luxury
- Train, flight, bus or self-drive
- Domestic, international or both

Taste

- Nature
- Beach
- Adventure
- Culture
- Food
- Spiritual
- Nightlife
- Shopping

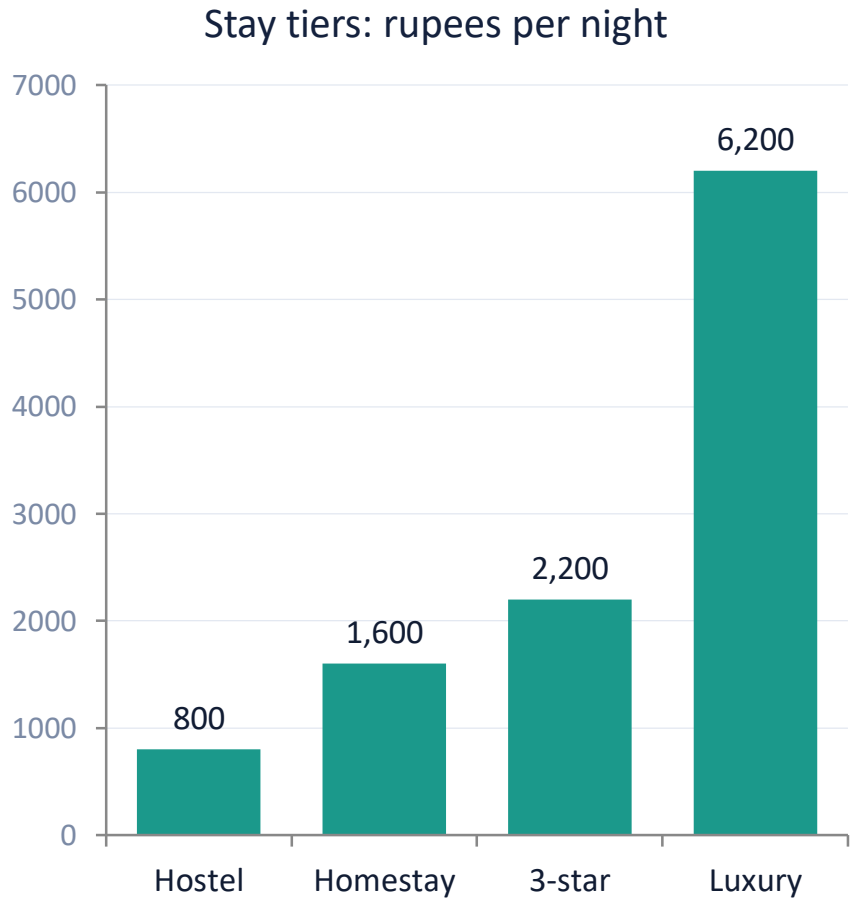
Validation: trip length and travellers must be positive; budget must be non-negative.

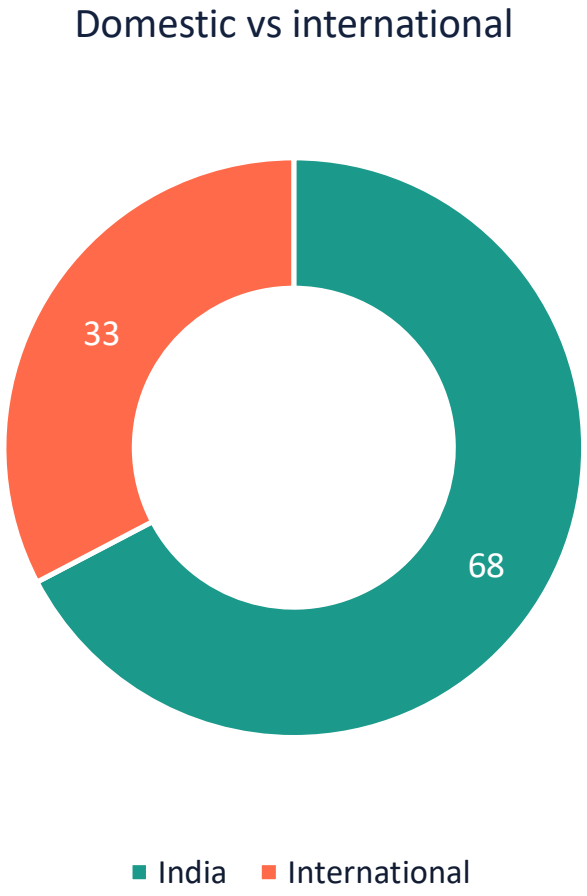
The knowledge base behind every decision

101

destinations across India and the world

Each record holds categories, tags, budget tier, daily cost, trip-length range, best months, suitable travellers and pace.





Eight rules score every destination

0

Travel scope

Domestic or international preference

-80 on mismatch

1

Budget feasibility

Budget vs estimated trip cost

+25 to -28

2

Interest match

Interests vs categories and tags

+14 per category

3

Season fit

Trip month vs best months

+18 / +6 / -10

4

Duration fit

Days vs ideal trip length

+15 to -20

5

Traveller type

Solo, couple, family or group

+12 / -5

6

Origin and transport

Proximity and flight needs

+10 / +5 / -22

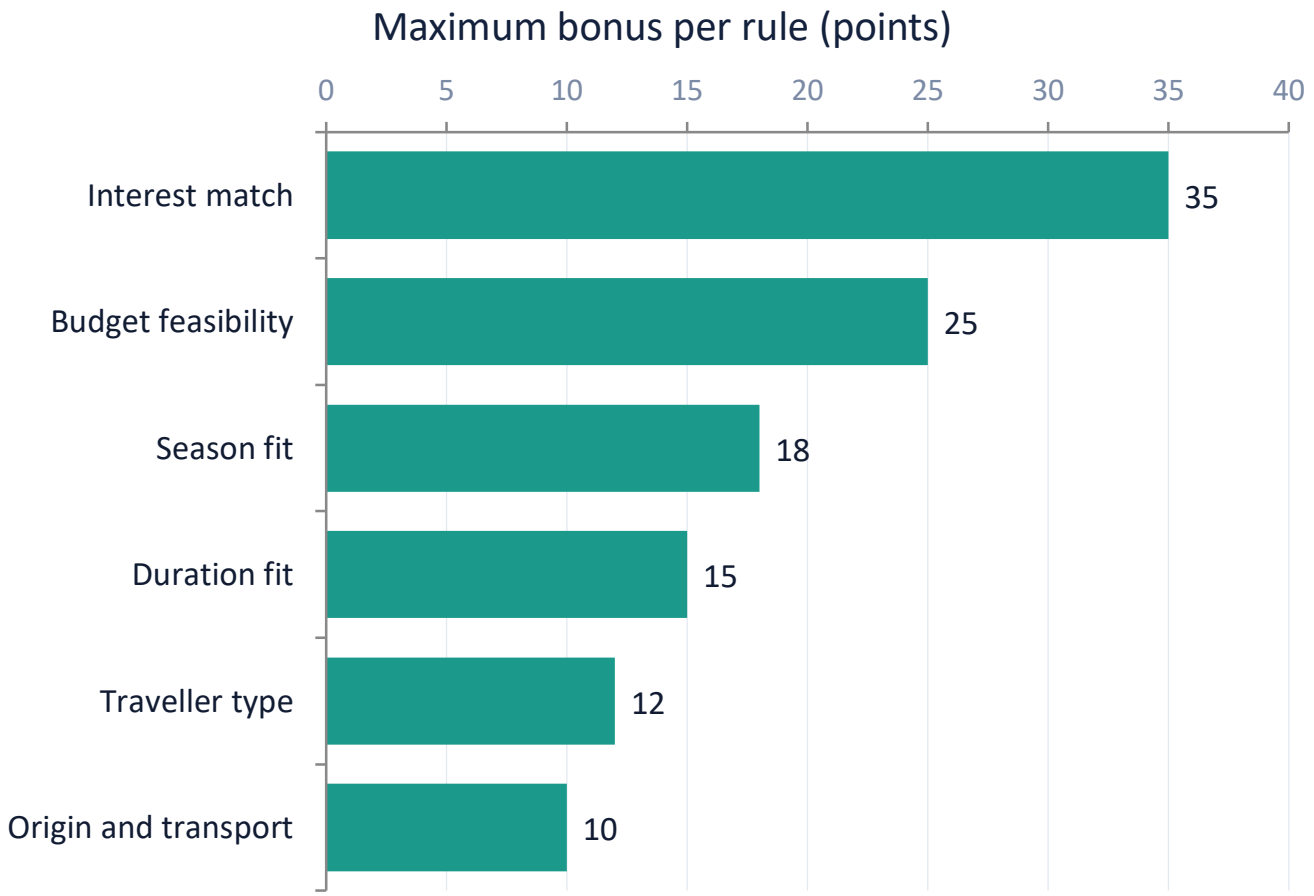
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Diversity re-ranking

Avoids repetitive top results

Penalty on repeats

From rule points to a ranked shortlist

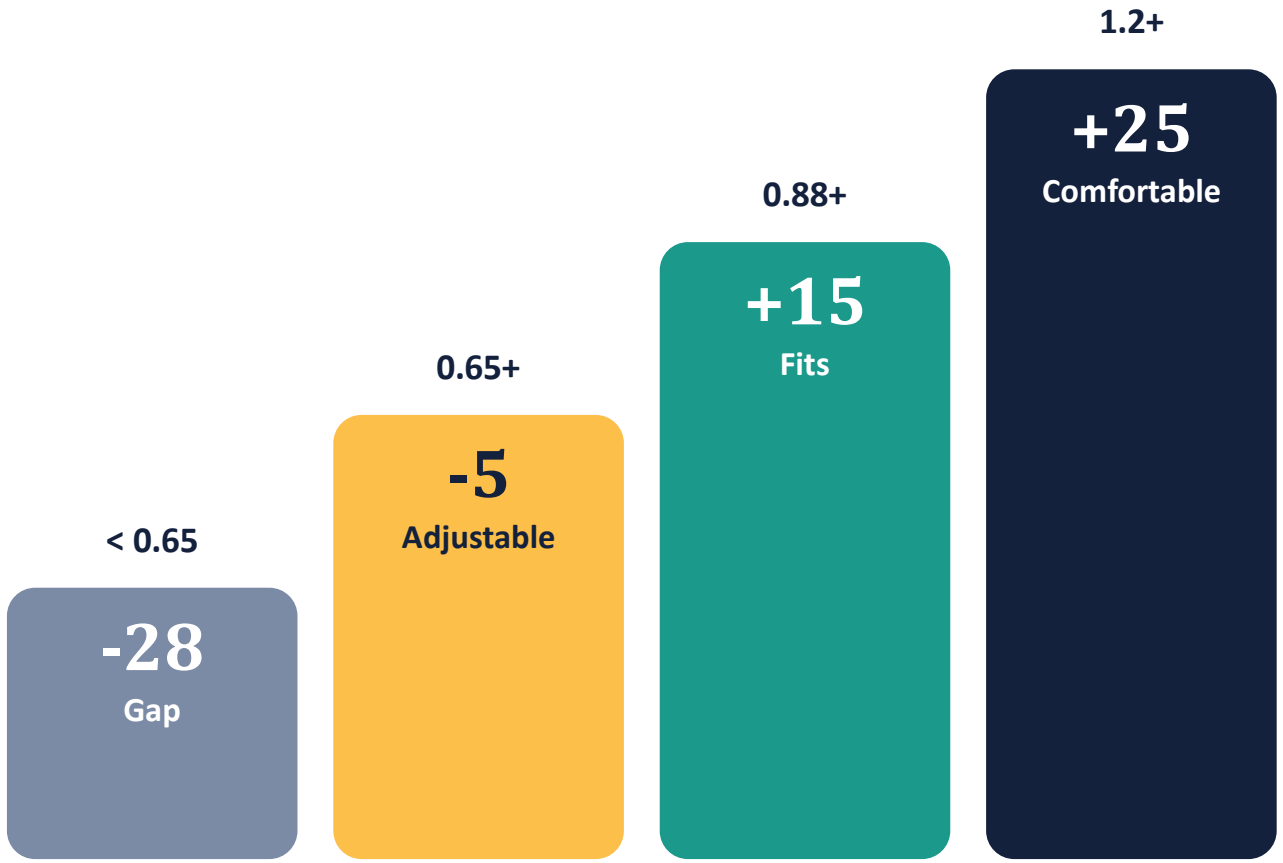


SCORE PIPELINE

1. Start every destination at 50 points
2. Add or subtract points from each rule
3. Floor the score at 10
4. Match % = score / 140, shown between 28% and 99%
5. Diversity pass trims repeated regions and categories

Budget feasibility: a four-step staircase

Budget ratio = your budget / estimated minimum cost



ESTIMATED MINIMUM COST

Transport + Stay + Ground + Food

Stay
nightly rate x days x rooms (one room per two travellers)

Ground
daily cost x days x travellers x tier multiplier

Food
Rs 750 x days x travellers

Transport
per-traveller fare; international flights costed by region

Itinerary pattern: three kinds of day, three slots each

	Day 1 Arrival and orientation	Middle days Iconic exploration	Last day Highlights and departure
Morning	Arrive and check in	Top attraction in best light	Highlights and souvenirs
Afternoon	First highlight	Local immersion	Final sightseeing
Evening	Relax and dine	Unwind, local flavour	Wrap up and depart

Illustrative pattern. Activities are spread across days to avoid overloading any one day; this is a suggested schedule, not an optimised route.

Every score comes with a receipt

rule log (illustrative entries)

PASS	**RULE_01_BUDGET_FEASIBILITY** Budget comfortably covers the estimated trip cost	+25
PASS	**RULE_03_SEASON_CLIMATE_MATCH** Trip month falls in the ideal season	+18
WARN	**RULE_04_DURATION_FIT** Trip is longer than the usual maximum stay	+4
FAIL	**RULE_00_TRAVEL_SCOPE** Destination is international but domestic was chosen	-80



Traceable

Each pick maps to an identifiable rule



Editable

Change a rule, not a trained model



Testable

Rules can be checked with defined test cases

Built as a working web application



Front end: React + Vite

Planner wizard, destination cards, itinerary dashboard, rule viewer



Logic: JavaScript rule engine

Scores destinations, allocates budget, builds the day-wise plan; no ML libraries



Back end: FastAPI + SQLite

Hashed-password sign-in, sessions, trip generation, budget and saved plans

API ENDPOINTS

```
POST /api/auth/register

POST /api/auth/login

POST /api/trips/recommend

POST /api/trips/generate

POST /api/budget/calculate

GET, POST, DELETE /api/plans
```

Test plan and honest limits

T1

History interest

Heritage sites ranked first

T2

Nature interest

Nature places recommended

T3

Limited budget

Low-cost options rank higher

T4

Short trip

Fits the daily time limit

T5

Over budget

Warning and cheaper options

T6

Invalid input

User asked to correct it

T7

No matching data

Clear no-data message

T8

Changed inputs

Plan regenerates

Measured results are not claimed in the paper; outcomes will be recorded after testing.



Limits

- Only stored destinations
- Prices can go stale
- No live hours, weather or traffic
- Hand-written rules
- Suggested, not optimal, routes

Arrival: what we built and where next

8
scoring rules
each one logged and explainable

101
destinations
68 in India, 33 abroad

0
training data needed
no ML model to train

NEXT STOPS



Maps and routing

Live weather, hours and prices

Multilingual support

PDF itineraries

Thank you. Questions?